

Manage TPM lockout

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Applies to:  [Windows 11](#),  [Windows 10](#),  [Windows Server 2022](#),  [Windows Server 2019](#),  [Windows Server 2016](#)

This article for the IT professional describes how to manage the lockout feature for the Trusted Platform Module (TPM) in Windows.

About TPM lockout

The TPM locks itself to prevent tampering or malicious attacks. TPM lockout often lasts for a variable amount of time or until the computer is turned off. While the TPM is in lockout mode, it generally returns an error message when it receives commands that require an authorization value. One exception is that the TPM always allows the owner at least one attempt to reset the TPM lockout when it is in lockout mode.

Windows takes ownership of the TPM ownership upon first boot. By default, Windows doesn't retain the TPM owner password.

In some cases, encryption keys are protected by a TPM by requiring a valid authorization value to access the key. A common example is configuring BitLocker Drive Encryption to use the TPM plus PIN key protector. In this scenario, the user must type the correct PIN during the boot process to access the volume encryption key protected by the TPM. To prevent malicious users or software from discovering authorization values, TPMs implement protection logic. The protection logic is designed to slow or stop responses from the TPM if it detects that an entity might be trying to guess authorization values.

TPM 2.0

TPM 2.0 devices have standardized lockout behavior which Windows configures. TPM 2.0 devices have a maximum count threshold and a healing time. Windows configures the maximum count to be 32 and the healing time to be 10 minutes. This configuration means that every continuous 10 minutes of powered on operation without an event causes the counter to decrease by 1.

If your TPM is in lockout mode or is responding slowly to commands, you can reset the lockout value by using the following procedures. Resetting the TPM lockout requires the TPM owner's authorization. This value is no longer retained by default starting with Windows 10 version 1607 and higher.

TPM 1.2

The industry standards from the Trusted Computing Group (TCG) specify that TPM manufacturers must implement some form of protection logic in TPM 1.2 and TPM 2.0 chips. TPM 1.2 devices implement different protection mechanisms and behavior. In general, the TPM chip takes exponentially longer to respond if incorrect authorization values are sent to the TPM. Some TPM chips might not store failed attempts over time. Other TPM chips might store every failed attempt indefinitely. Therefore, some users might experience increasingly longer delays when they mistype an authorization value that is sent to the TPM. These delays can prevent them from using the TPM for some time.

Reset the TPM lockout by using the TPM MMC

ⓘ Note

This procedure is only available if you have configured Windows to retain the TPM Owner Password. By default, this password isn't available in Windows 10 starting with version 1607 and higher.

The following procedure explains the steps to reset the TPM lockout by using the TPM MMC.

Reset the TPM lockout

1. Open the TPM MMC (tpm.msc).
2. In the **Action** pane, select **Reset TPM Lockout** to start the Reset TPM Lockout Wizard.
3. Choose one of the following methods to enter the TPM owner password:
 - If you saved your TPM owner password to a .tpm file, select **I have the owner password file**, and then type the path to the file, or select **Browse** to navigate to the file location.
 - If you want to manually enter your TPM owner password, select **I want to enter the owner password**, and then type the password in the text box provided.

ⓘ Note

If you enabled BitLocker and your TPM at the same time, and you printed your BitLocker recovery password when you turned on BitLocker, your TPM owner password may have printed with it.

Use Group Policy to manage TPM lockout settings

The TPM Group Policy settings in the following list are located at:

Computer Configuration > Administrative Templates > System > Trusted Platform Module Services

- [Standard User Lockout Duration](#)

This policy setting allows you to manage the duration in minutes for counting standard user authorization failures for TPM commands that require authorization. An authorization failure occurs each time a user sends a command to the TPM and receives an error message that indicates an authorization failure occurred. Authorization failures that are older than the duration you set are ignored. If the number of TPM commands with an authorization failure within the lockout duration equals a threshold, the user is prevented from sending commands to the TPM that require authorization.

- [Standard User Individual Lockout Threshold](#)

This policy setting allows you to manage the maximum number of authorization failures for the TPM for each user. This value is the maximum number of authorization failures that each user can have before the user isn't allowed to send commands to the TPM that require authorization. If the number of authorization failures equals the duration that is set for the policy setting, the user is prevented from sending commands to the TPM that require authorization.

- [Standard User Total Lockout Threshold](#)

This policy setting allows you to manage the maximum number of authorization failures for the TPM for all standard users. If the total number of authorization failures for all users equals the duration that is set for the policy, all users are prevented from sending commands to the TPM that require authorization.

For information about mitigating dictionary attacks that use the lockout settings, see [TPM fundamentals](#).

Use the TPM cmdlets

You can manage the TPM using Windows PowerShell. For details, see [TPM Cmdlets in Windows PowerShell](#).